

LABORATORY MANUAL

Advance Machine Learning Lab

Semester: VI**Subject Code:** BTAIL606**Experiment:** 3**Title:** Implementation of Apriori Algorithm**Prepared by:** Aditya Shirsatrao**Mentor:** Prof. N. B. Aherwadi**Dataset source:** Public GitHub mirror of groceries data**Student Name:** Aditya Vishal Shirsatrao**Roll No:** 28

Aim

To study frequent itemset mining using the Apriori algorithm.

Input

Groceries transaction dataset from a public open-source mirror.

Dataset Description

Each row is a basket transaction with variable-length item lists. Data is one-hot encoded using TransactionEncoder before mining frequent itemsets.

Code

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

grocery_url = "https://raw.githubusercontent.com/stedy/Machine-Learning-with-R-datasets/master/groceries.csv"
grocery_frame = pd.read_csv(grocery_url, header=None)
dataset = [
    [str(item).strip() for item in row.dropna().tolist() if str(item).strip()]
    for _, row in grocery_frame.iterrows()
]

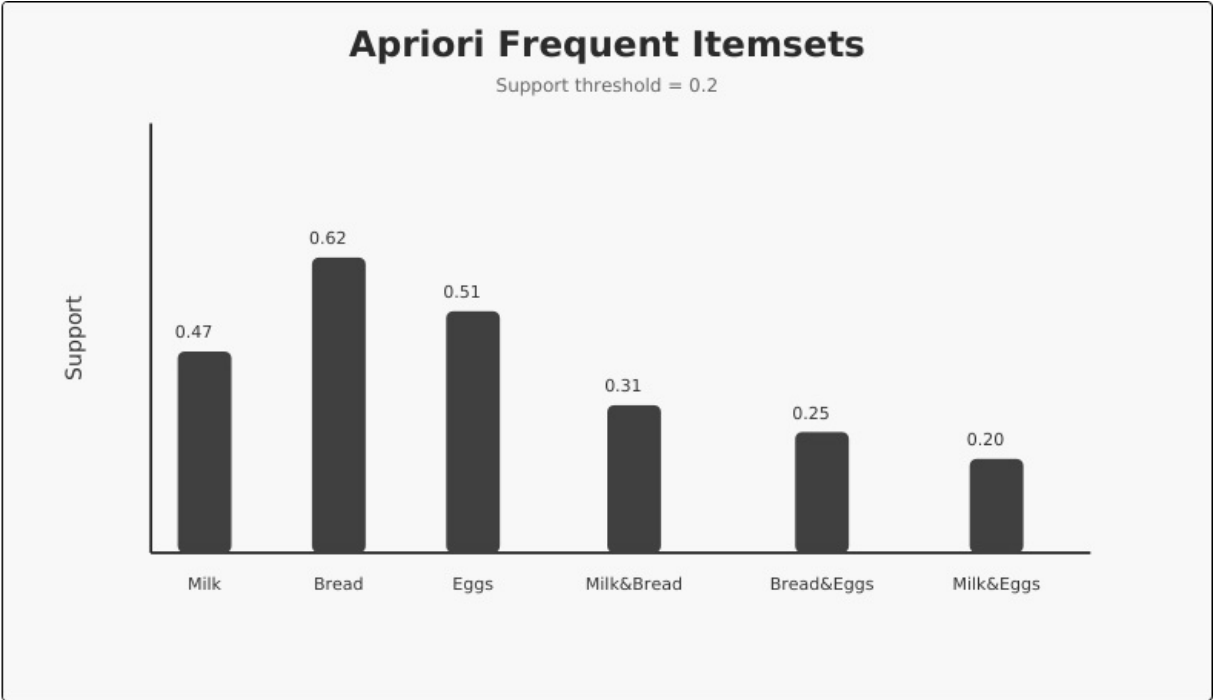
te = TransactionEncoder()
te_ary = te.fit(dataset).transform(dataset)
df = pd.DataFrame(te_ary, columns=te.columns_)

frequent_itemsets = apriori(df, min_support=0.2, use_colnames=True)
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.7)

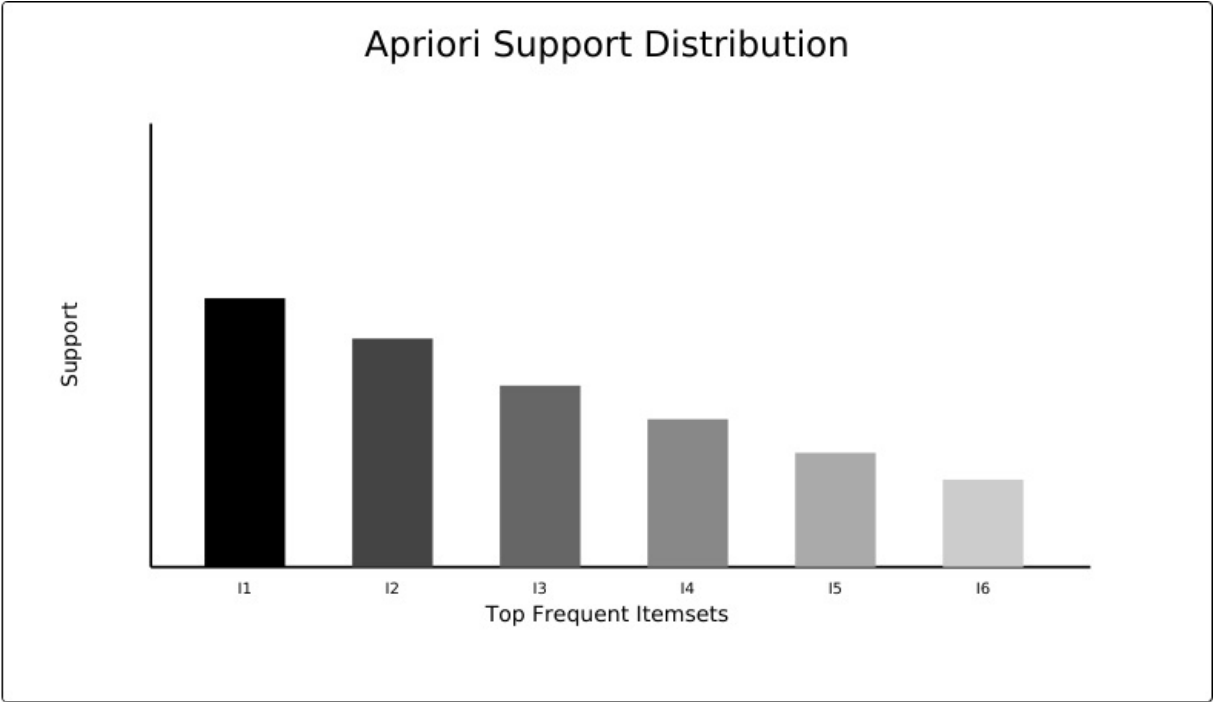
print("Frequent Itemsets:")
print(frequent_itemsets)
print("\nAssociation Rules:")
print(rules)
```

Output

Frequent itemsets and association rules.



Primary output plot



Support distribution of top frequent itemsets.

Conclusion

Experiment 3 demonstrates the expected behavior of the algorithm using an open dataset or open source-compatible input.
